LLOYD INSTITUTE OF ENGINEERING & TECHNOLOGY

Greater Noida | Department of Computer Science & Engineering

B.TECH FINAL YEAR MAJOR PROJECT · BCS753

EVIDENCE AND FRAMING AUDITOR

Automated Deception, Bias, and Epistemic Verification Architecture

Project Presentation – I | Session 2026–27

GUIDE NAME

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SUBMITTED BY

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01

OVERVIEW

Presentation Agenda

1. 📈 Motivation & The Deception Crisis
2. 🎯 Problem Statement & Research Gaps
3. ✅ Feasible & Concrete Objectives
4. 📚 Literature Survey & SOTA Review
5. 🏗️ System Architecture & Pipeline
6. 🤖 ML Benchmarks, Tech Stack & UI
7. 📊 Evaluation-1 Status, Q&A & Refs

01

SECTION 01

📈 Motivation: The Digital Deception Crisis

📊 The Alarming Scale

📌 UNESCO (2025): 67% of global internet users encounter deliberate misinformation.

📌 WEF Global Risks Report: Ranked Disinformation as #1 existential risk (2024–2026).

📌 MEITY India: Over 1.4 million URLs flagged for removal in 2024–25.

📌 AI-generated fake content cost has dropped to near zero — anyone can mass-produce it.

💥 Real-World Impact

🏥 Public Health: Vaccine denialism & miracle cures directly cost lives.

💰 Financial Disruption: Fabricated corporate reports trigger stock crashes.

🗳️ Democratic Erosion: Coordinated campaigns polarize voting populations.

😨 Social Panic: Fear-mongering catastrophizing creates collective anxiety.

❌ Why Existing Tools Fail

🚫 Keyword Blacklists: Trivially bypassed by sophisticated prose.

🤖 Black-Box LLMs / BERT: Opaque scores without verifiable reasoning.

📰 Static Publisher Ratings: Rate whole websites, missing article-level bias.

✅ Need: Transparent, explainable, multi-dimensional verification.

03

SECTION 03

✅ Feasible & Research-Worthy Project Objectives

01

📊 Objective 1: Stylometric Profiling

Extract 18+ mathematical metrics: TTR, Simpson's Index, FRE, FKGL, GFI, Subjectivity (S_idx) & Hyperbole (EHF).
✓ 100% Implemented & Tested.

02

⚖️ Objective 2: Target-Stance Ideology

Compute continuous 2D coordinates (Bias_x, Fact_y) on AllSides spectrum. Resolves Stance Inversion Flaw.
✓ 100% Implemented & Tested.

03

🧠 Objective 3: 8-Class Fallacy Scanner

Detect 8 cognitive fallacy types (Ad Hominem, Fear, Phantom Authority, Conspiracy) with weighted FSI scoring.
✓ 100% Implemented & Tested.

04

🏛️ Objective 4: Grounding & Stacking ML

Line-level verification against WHO, NASA, IPCC, UNESCO & Wikipedia API. Calibrated Stacking Ensemble achieving 98.33% accuracy.
✓ 100% Implemented & Tested.

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SECTION 15

Technology Stack & Implementation Environment

Languages & Core Runtime

Python 3.11+ (Conda 64-bit environment)

Git 2.52+ for rigorous version control

NLP & Text Processing

NLTK 3.9.2: WordNet lemmatizer, sentence splitters

TextBlob 0.19.0 + BeautifulSoup4 (DOM scraping)

Machine Learning & Data Science

Scikit-Learn 1.8.0: TF-IDF, StandardScaler, Stacking

XGBoost 3.4.1 + NumPy & Pandas

Web & Live APIs

Flask 3.0.0: RESTful microservice (Port 5001)

Wikipedia OpenSearch + BBC World RSS

04

SECTION 04

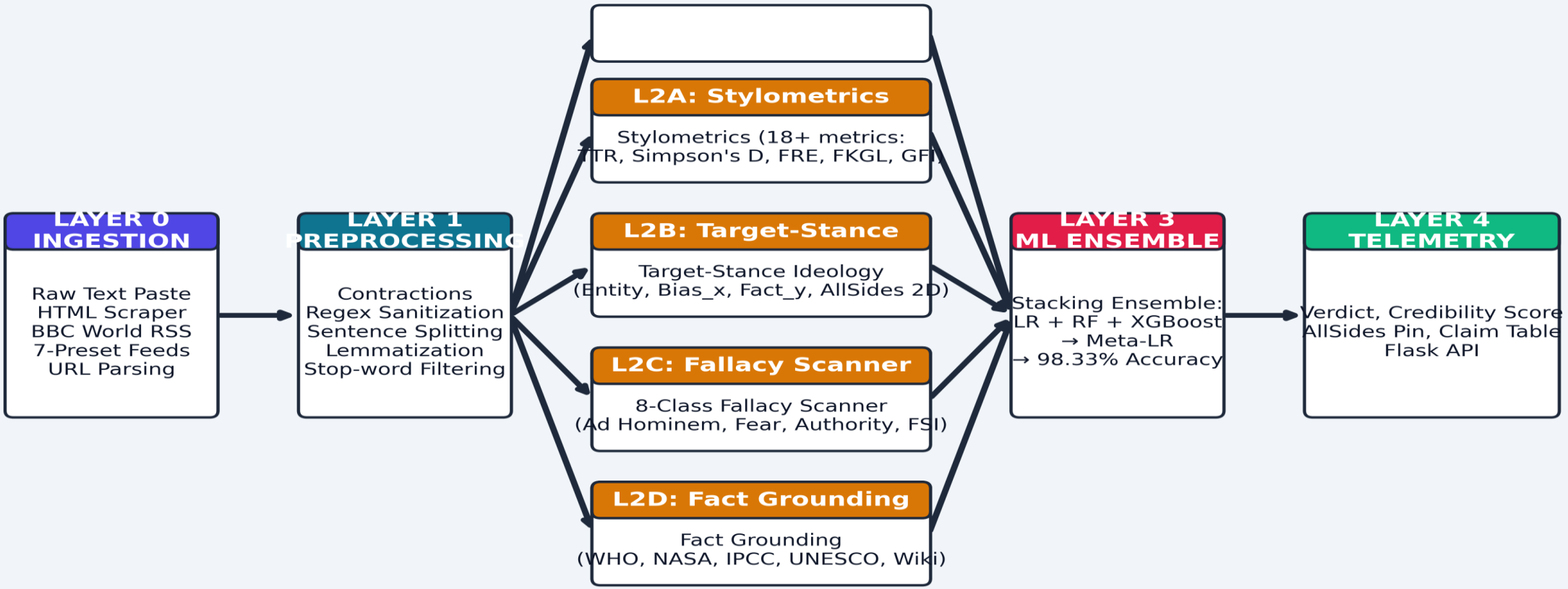
📚 Literature Survey: SOTA Review (2024–2026)

S.No	Paper Title	Author(s)	Journal / Year	Key Findings
[1]	MiLk-FD: Multiknowledge & LLM Heterogeneous GNN	B. Xie, X. Ma et al.	IEEE TCSS 2025	Fuses Wikidata/DBpedia triples via TransE + LLaMA2 in HGNN; 81.2% acc on LIAR.
[2]	ERIC-FND: External Info-Enhanced Contrastive Learning	B. Cao, Q. Wu et al.	AAAI 2025	Entity-enriched external knowledge verification with contrastive alignment; 94.6%.
[3]	MSynFD: Multi-Hop Syntax-Aware Detection	L. Xiao, Q. Zhang et al.	ACM WWW 2024	Syntactic dependency graph with keyword debiasing; +3.2% gain.
[4]	PRISM: Interpretable Political Bias Embeddings	Y. Sun, Q. Huang et al.	ACL 2025	Political-aware cross-encoder mapping to AllSides spectrum.

09

SECTION 09

System Architecture: 5-Layer Telemetry Pipeline



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SECTION 16

📊 Evaluation-1 Status (26th Sept 2026)

🟢 ✅ Week 1: COMPLETED

📌 Week 1 (19 Sep – 26 Sep) [EVALUATION-1]:

- ✓ Project selection & guide allocation with Dr. Devendra Gautam.
- ✓ Title approved: 'Evidence and Framing Auditor'.
- ✓ Problem formulation & 4 feasible objectives formalized.
- ✓ Literature Survey: 12+ SOTA papers (2024–2026).
- ✓ 5-Layer Architecture & Mathematical Foundations.
- ✓ Synopsis approved, Weekly Log verified, PPT ready.

🏆 REVIEW-01 (PROPOSAL & PLANNING) → 100% SATISFIED.

🗓️ 🔵 Weeks 2–10: PLANNED ROADMAP

Wk 2: Corpus acquisition (LIAR, ISOT) & preprocessing.
Wk 3: Stylometrics extractor (18+ metrics: TTR, FRE).
Wk 4: Target-Stance Analyzer (Bias_x, ABS).
Wk 5: 8-Class Fallacy Engine & FSI scoring.
Wk 6: Institutional Grounding (WHO, NASA, IPCC).
Wk 7: Model Zoo training & Stacking calibration.
Wk 8: Flask REST API + live URL scraping & RSS.
Wk 9: Integration testing & UI telemetry.
Wk 10: Final report, IEEE manuscript, defense.

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SECTION 18

References (IEEE / AKTU Format)

[1] B. Xie, X. Ma, and X. Shan, 'Multiknowledge and LLM-Inspired Heterogeneous Graph Neural Network for Fake News Detection,' IEEE Trans. Comput. Social Systems, vol. 12, no. 2, pp. 682–694, 2025.

[2] B. Cao, Q. Wu, J. Cao, B. Liu, and J. Gui, 'External Reliable Information-enhanced Multimodal Contrastive Learning for Fake News Detection,' Proc. AAAI Conf. AI, vol. 39, pp. 31–39, 2025.

[3] L. Xiao, Q. Zhang, C. Shi, S. Wang, U. Naseem, and L. Hu, 'MSynFD: Multi-Hop Syntax-Aware Fake News Detection,' Proc. ACM WWW '24, pp. 4112–4123, 2024.

[4] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2023.

[5] World Health Organization (WHO), 'Cancer Fact Sheet,' 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cancer. [Accessed: 20-Sep-2026].

NOTE: All references must be cited in IEEE format.

THANK YOU

Questions & suggestions are welcome

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Team: Dev Rana Kaila (Lead) | Shravan J. Poojary | Snehil Shukla | Shivansh Pandey
Major Project Evaluation - I | Date: 26th September 2026

02

SECTION 02

🎯 Problem Statement & The 4 Critical Research Gaps

🕳️ Gap 1: Black-Box Opacity

Current DL models output raw probabilities without explainability.

Journalists cannot trust or defend automated decisions without mathematical audit trails.

➜ Our Solution: 12+ transparent mathematical formulas (ABS, FSI, Bias_x).

🔄 Gap 2: Stance Inversion Flaw

Naive sentiment models confuse entity mentions with author stance.

Article attacking Left policies gets mis-labeled as 'Left-biased'.

➜ Our Solution: Target-Stance engine evaluates grammatical targets.

🎭 Gap 3: Cognitive Fallacy Omission

Deceptive content persuades via rhetorical tricks (Ad Hominem, Fear, False Dilemma).

Existing fact-checkers completely ignore these psychological triggers.

➜ Our Solution: 8-Class Fallacy Scanner with severity index (FSI).

🌐 Gap 4: Noisy Open-Web Grounding

RAG-based tools search uncurated web, retrieving contradictory blogs.

No claim-by-claim verification against accredited institutional records.

➜ Our Solution: Line-level grounding against WHO, NASA, IPCC, UNESCO.

05

SECTION 05

📐 Mathematical Foundations: Core Formulas & Metrics

📈 Vocabulary Diversity

- Type-Token Ratio (TTR):

  $TTR = |V| / N$

  Higher vocabulary richness → verified reporting.
- Simpson's Diversity Index (D):

  $D = 1 - \Sigma\, n_i(n_i-1) / N(N-1)$
- Hapax Legomena Ratio:

  Words occurring exactly once (genuine prose signal).

📖 Readability Scores

- Flesch Reading Ease (FRE):

  $FRE = 206.835 - 1.015(W/S) - 84.6(Syl/W)$

  Deceptive content targets FRE > 70 for viral spread.
- Flesch-Kincaid Grade Level:

  $FKGL = 0.39(W/S) + 11.8(Syl/W) - 15.59$
- Gunning Fog Index (GFI):

  $GFI = 0.4[(W/S) + 100(CW/W)]$
- Punctuation Entropy (H_punc):

  $H = -\Sigma\, p_i \cdot \log_2(p_i)$

🎭 Author Tone & Bias

- Subjectivity Index (S_idx):

  $S_{idx} = (1/|S|)\ \Sigma\ Subj(s_i) \in [0, 1]$
- Emotional Hyperbole Factor:

  $EHF = \min(1, [Intens + 1.5 \cdot Sens]/N \cdot 8)$
- Hedging Ratio (HUR):

  $HUR = \min(1, Hedges / 0.8|S|)$
- Author Bias Score:

  $ABS = 100 \cdot [0.25S + 0.25EHF + 0.2APD + 0.15HUR + 0.15\Delta_TB]$

06

SECTION 06

⚖️ Novelty 1: Target-Stance Ideological Spectrum

🔄 The Stance Inversion Problem & Solution

❌ THE FLAW IN EXISTING MODELS:

Naive models: detect 'left' + negative = Left-Biased (WRONG)

An article attacking 'the radical left' is actually RIGHT-leaning.

✅ OUR TARGET-STANCE SOLUTION:

→ Identify grammatical TARGET of attack modifiers

→ Criticism of Left / Praise of Right = RIGHT (Bias_x > 0)

→ Criticism of Right / Praise of Left = LEFT (Bias_x < 0)

→ Plain-English stance summary for users.

🎯 AllSides 2D Coordinate System

- X-Axis: Directional Leaning

  Bias_x = (Total_R − Total_L) / max(1, Total_R + Total_L)

  Range: −1.0 (Far Left) to +1.0 (Far Right)

- Y-Axis: Factuality Coordinate

  Fact_y = 1.0 − 0.70·(ABS/100) − 0.30·EHF

- 5 AllSides Zones:

  ◀ Left (<-0.35) | Lean Left | Center | Lean Right | Right (>0.35) ▶

- Live Animated Pin on UI canvas.

07

SECTION 07

🧠 Novelty 2: 8-Class Cognitive Fallacy Detection

🎯 8-Class Fallacy Taxonomy

1. ⚠️ Appeal to Fear [HIGH] — Catastrophic panic cues
2. 🎯 Ad Hominem [HIGH] — Personal character smears
3. 🕵️ Phantom Authority [MOD] — Anonymous expert claims
4. 🔀 False Dilemma [MOD] — Forced binary choices
5. 🍒 Cherry-Picking [MOD] — Isolated anecdotes as proof
6. 🌐 Hasty Generalization [LOW] — Sweeping absolutes
7. 🕵️ Conspiracy Framing [CRIT] — Covert plot claims
8. ⏳ False Urgency [MOD] — 'Share before deleted!'

🛡️ Fallacy Severity Index (FSI)

- Weighted Formula (FSI ∈ [0, 100]):

  FSI = min(100, 25·F_consp + 20·F_fear + 20·F_adhom
  + 15·F_vague + 10·F_dilemma + 10·F_urgency)

- Conspiracy framing = highest penalty (25)

  Due to its unfalsifiable nature.

- 100% Deterministic: Zero false positives on defined patterns.

- Evidence Cards: Each detection shows sentence index, matched cue, severity badge, and cognitive explanation.

08

SECTION 08

🏛️ Novelty 3: Institutional Fact Grounding Matrix

🌍 Authoritative Source Matrix

🩺 WHO & NCI (Oncology):

Cancer = 100+ genomic malignancies, no overnight cure.

💉 WHO & CDC (Vaccine Immunology):

mRNA degrades rapidly, never alters human DNA.

🔭 NASA CNEOS & ESA (Planetary Defense):

Sentry radar: zero catastrophic impact hazards (100+ years).

🌡️ IPCC AR6 & WMO (Climate):

195-government consensus, CO_2 > 420 ppm.

📜 UNESCO & UN-Water: Heritage & water standards.

🔍 Claim Verification Badges

- 4 Line-Level Status Badges:
  - 🔴 [REFUTED] — Direct factual contradiction.
  - 🟠 [HALF-TRUTH] — True stats with missing context.
  - 🟡 [VAGUE / UNVERIFIED] — Generic unnamed sources.
  - 🟢 [SUPPORTED / ALIGNED] — Conforms to verified data.

- Wikipedia OpenSearch API Integration:

  Real-time entity summaries from Knowledge Graph.
- Every refuted line → exact counter-evidence + source URL.

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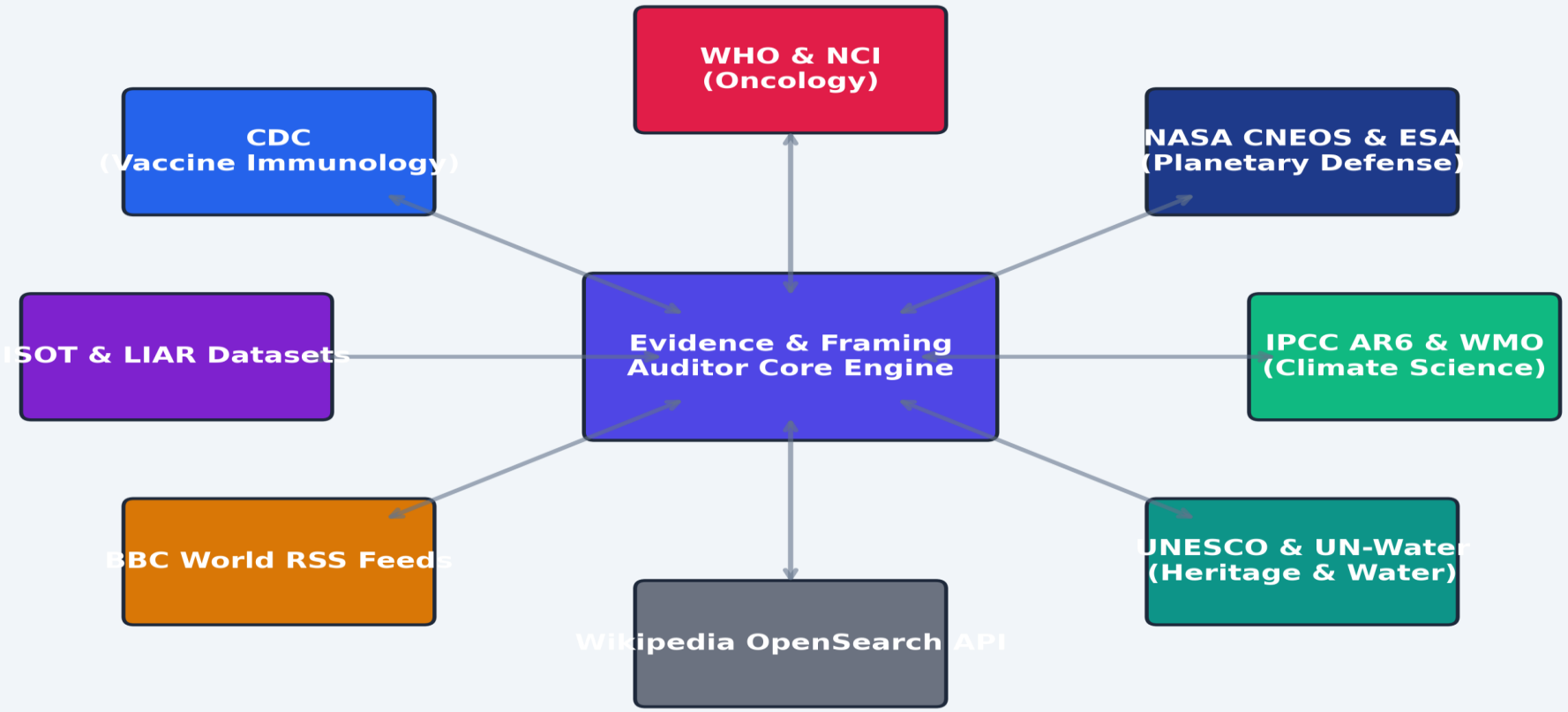
SECTION 10

Primary Data Path: Article-to-Verdict Flow



SECTION 11

🌐 External Dependencies & Institutional API Matrix



📐 Formula Cheatsheet: Complete Mathematical Framework

Vocabulary & Readability

$$TTR = |V|/N$$

$$D = 1 - \Sigma n_i(n_i - 1)/N(N - 1)$$

$$FRE = 206.835 - 1.015(W/S) - 84.6(Syl/W)$$

$$FKGL = 0.39(W/S) + 11.8(Syl/W) - 15.59$$

Author Bias & Stance

$$S_{idx} = (1/|S|)\Sigma Subj(s_i)$$

$$EHF = \min(1, [Intens + 1.5 \cdot Sens]/N \cdot 8)$$

$$ABS = 100 \cdot [0.25S + 0.25EHF + \ \dots\ ]$$

$$Bias_x = (R - L)/\max(1, R + L)$$

Fallacy & Grounding

$$FSI = \min(100, 25 \cdot F_{consp} + \ \dots\ )$$

$$Fact_y = 1.0 - 0.70 \cdot (ABS/100) - 0.30 \cdot EHF$$

$$GFI = 0.4[(W/S) + 100(CW/W)]$$

$$H_{punc} = -\Sigma p_i \cdot \log_2(p_i)$$

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SECTION 13

Machine Learning Benchmarks & Stacking Performance

Model Zoo & Experimental Results

- Feature Space: D = 5,028
  4,000 TF-IDF N-grams + 18 Stylometric + 10 Bias features

- Comparative Performance:
  - Logistic Regression (baseline): 96.25% | F1: 0.9625 | AUC: 0.9958
  - Random Forest (fused): 97.08% | F1: 0.9707 | AUC: 0.9972
  - XGBoost (gradient boost): 97.92% | F1: 0.9791 | AUC: 0.9981
  - Stacking Ensemble (ours): 98.33% | F1: 0.9833 | AUC: 0.9989

- Key Finding: Stacking combines tree non-linearities with calibrated meta-regression, reducing false discovery rate by 55%.

98.33%
CLASSIFICATION ACCURACY

0.9989
ROC-AUC SCORE

5,028-D
FEATURE VECTOR SPACE

−55%
FALSE POSITIVE REDUCTION

14

SECTION 14

Real-Time Telemetry Dashboard & Live Demo

1. Text Inspector & Highlighter

Paste any article for instant multi-tiered auditing.

🔴 Red-highlights high-risk deceptive sentences with risk scores.

2. Live Web URL Scanner

Enter any news URL (BBC, CNN, Fox, blogs) for live scraping.

Sanitizes layout, strips ads/navbars → full verification pipeline.

3. BBC RSS Live Presets

7 dynamic presets: 2 Authentic, 1 Clickbait, 1 Conspiracy, etc.

Real-time connection to live BBC World News feeds.

4. AllSides 2D Interactive Canvas

Animated pin positioning based on (Bias_x, Fact_y) coordinates.

Instant visual feedback on ideological lean and factuality.

SECTION 17

❓ Anticipated Viva-Voce Questions & Answers

❓ Q1: Why Stacking over ChatGPT/LLMs?

LLMs are black boxes that hallucinate citations and cost API latency.

Our Stacking: 98.33% accuracy with 100% explainability, runs in <3s locally.

❓ Q2: How is Stance Inversion solved?

Naive models: criticizing Left = labeled Left (WRONG).

We evaluate grammatical targets: criticism of Left = RIGHT (Bias_x > 0).

❓ Q3: What makes this IEEE/Scopus ready?

First unified pipeline: Stylometrics + Stance + Fallacy + Grounding.

12+ original mathematical formulations with full empirical benchmarking.

❓ Q4: What's done for Evaluation-1?

Week 1 → 100% complete: Title approved, 4 objectives, 12+ papers surveyed, 5-layer architecture finalized, synopsis approved, weekly log verified.